LECTURE 1

WHAT IS THE GREATEST TRADE EVER?

GREAT TRADES

International Business Times' Top Ten Greatest Trades

Rank	Trader	Position	Year
1	John Paulson	Against subprime	2008
2	Jesse Livermore	Against US equity	1929
3	John Templeton	Long Japan	1960s–1990s
4	George Soros	Against UK pound	1992
5	Paul Tudor Jones	Short Black Monday	1987
6	Andrew Hall	Long oil	2003
7	David Tepper	Long financials	2009
8	Jim Chanos	Short Enron	2001
9	Jim Rogers	Long commodities	1990s–
10	Louis Bacon	Long oil, short equity	1990

GREAT TRADES

At least half are **derivatives** trades

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PAULSON'S BET AGAINST SUBPRIME

PAULSON'S SUBPRIME POSITION

Paulson's hedge fund specializes in risk arbitrage

Risk arbitrage is speculation on mergers and acquisitions

He is an investment banker, not a macro-investor

But in 2007 he was convinced there was a bubble in US housing

How can you bet on this view?

PAULSON'S DERIVATIVES POSITION

You cannot short sell real estate

You cannot even short sell mortgages

But mortgages bundled into mortgage-backed securities (MBS)

You might be able to short MBS

But do you know another way to trade on Paulson's view?

PAULSON'S DERIVATIVES SQUARED POSITION

A credit default swap (CDS) pays off when a bond defaults

CDS typically thought of as insurance products

If long corporate bond, CDS insure against default

- Like fire insurance when you own a house
- When asset (house) not valuable, insurance contract pays off

But you can buy CDS without being long a bond

- Like fire insurance on neighbor's house

PAULSON'S PROFITS

Paulson and Co. made 15 billion USD on CDS on subprime MBS

Paulson took home almost five billion USD himself

- BTW: five billion seconds is about 159 years

Fund fell post crisis due to another macro derivatives position

- Long position in gold futures

WHO WAS ON THE OTHER SIDE OF THE TRADE?

Goldman Sachs marketed the MBS Paulson bet against

Paulson was involved in the structuring

SEC charged Goldman, saying it mislead investors

Goldman paid 550 million USD to settle

- At the time the biggest settlement in Wall Street history

AIG's insolvency was the result of writing CDS before crisis

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Be warned! Derivatives are dangerous

SOROS BREAKS THE BANK OF ENGLAND

BACKGROUND

- Currency markets are largest markets by trading volume
  - Daily volume of about four trillion USD
  - Global goods and services trade about two percent FX trade
- European exchange rates unstable after Bretton Woods ended
- To stabilise FX, created European exchange rate mechanism
  - ERM gave currencies 2.5 percent band for FX
  - Bands set with respect to German mark
  - BTW: Portugal, Italy, and Spain had wider bands (familiar?)
  - Greece joined the ERM II in 2001 making the PIS the PIGS

SEEDS OF CRISIS

Since FX pegged to the mark, interest rates not independent

German reunification in 1990 caused huge fiscal stimulus

Germany raised interest rates to cool overheating economy

Other countries had to raise rates, following to stay in FX bands

Rate raises not based on fundamentals, thus costly to maintain

SOROS'S VIEW

Soros believed that the UK would not maintain rates

It was too costly for the UK to pay such high rates

Soros predicted UK would exit ERM

So it could stop promising these rates

And in so doing let the pound devalue

How can you bet on this view?

FORWARD CONTRACTS

To bet against a currency, use forward contracts

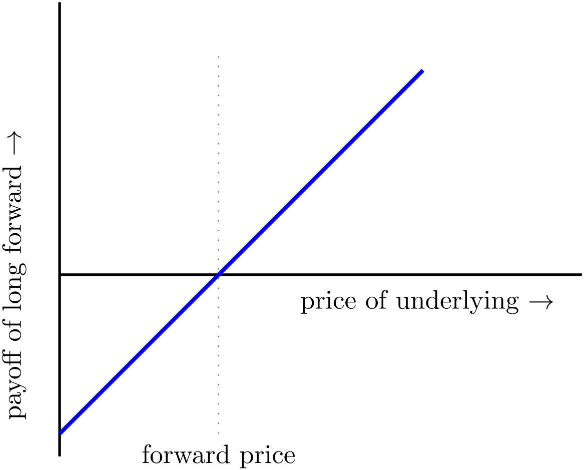
A forward contract on an underlying asset:

An agreement to buy or sell at a set price at a set time

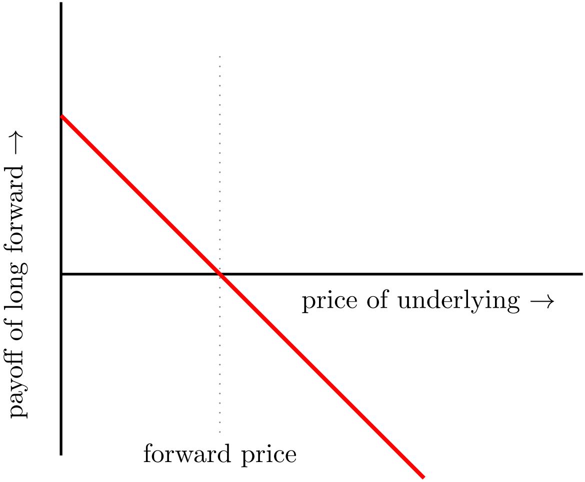
The agreed buyer is "long" and the agreed seller is "short"

Soros took a short position in forwards on the pound

LONG FORWARD PAYOFF DIAGRAM



SHORT FORWARD PAYOFF DIAGRAM



BLACK WEDNESDAY

On September 16, 1992

Bank of England lent at 12 percent, promising increase to 15

Soros and other speculators continued to short forwards

Downward price pressure led pound to close below ERM floor

UK estimated losses from defending the pound at 3.3 billion GBP

Soros made over one billion GBP in the speculative attack

ARE DERIVATIVES JUST FOR SPECULATION?

BEYOND SPECULATION

Sixty percent of corporates (non-financial firms) use derivatives

They use them to manage or <u>hedge</u> risk

Risk management is the main reason to use derivatives

How?

IBM AND WORLD BANK SWAP (1981)

IBM'S BALANCE SHEET MISMATCH

US rates were very high c. 1980 (17 percent)

Rates lower in Switzerland and West Germany (8 and 12 percent)

IBM had raised funds in Europe to take advantage of low rates

But IBM makes dollar investments

IBM had dollars on the left-hand side of its balance sheet

but European currencies on the right-hand side

This currency mismatch exposed IBM to FX risk

MEANWHILE, AT THE WORLD BANK

The World Bank lends to poor countries

Its lending rates are fixed by its cost of funding

Thus, the World Bank wants lowest funding rate globally

Had reached borrowing limits in Switzerland and West Germany

DOUBLE COINCIDENCE OF WANTS

IBM and the World Bank had a double coincidence of wants

IBM had too much Swiss and West German debt

The World Bank wanted more Swiss and German debt

The solution? A derivative called a currency swap

World Bank issued debt in US

entered offsetting swap agreement with IBM

Roughly: IBM made World Bank's USD coupon payments

and World Bank made IBM's coupon payments in Europe

Actually: exchanged money contingent on coupons

OLDER DERIVATIVES?

IBM-World Bank swap was the beginning of currency swaps

Notional value of currency swaps is now 10 trillion USD

But not all derivatives developed so recently...

THALES' BET ON (OLIVE) OIL

THALES

Thales of Miletus was a preSocratic philosopher from Anatolia

Called the Father of Science and the first mathematician

But in his time (c. 600 BC) he was reproached for his poverty

In response to criticism, Thales used his little money to speculate

With his knowledge of the stars, he forecast plentiful olive harvest

He needed to take a position on the olive harvest in the future

He developed “financial device...of universal application”

THALES' TRADE

Thales approached the operators of all the olive presses in Miletus

Paid upfront premium for the right to operate them in the future

After the plentiful harvest, all the presses were in high demand

Thales "let them out at any rate which he pleased"

He made a "quantity of money"

and "showed the world that philosophers can be rich if they like"

THALES' CONTRACTS

Thales' contracts had the following features

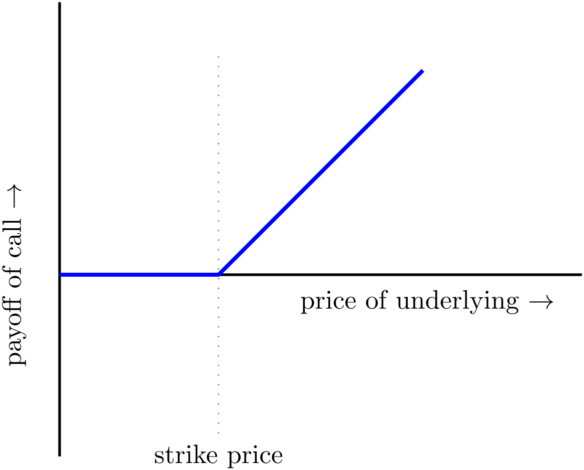
- Bought the right not the obligation to operate the presses
- Paid a premium upfront for the right

In modern parlance, this is an <u>option</u>

- Specifically, a <u>long position in a call option</u>

Derivatives are not so new!

(LONG) CALL OPTION PAYOFF DIAGRAM



RECAP

LECTURE 1: RECAP

Paulson's bet against subprime (CDS)

Soro's bet against the pound (currency futures)

IBM World Bank swap (swaps)

Thales' bet on oil (options)

FIN